Schur convexity and positive definiteness of the even degree complete homogeneous symmetric polynomials

Terence Tao · 6 August 2017

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The complete homogeneous symmetric polynomial $h_d(x_1, ..., x_n)$ of n variables $x_1, ..., x_n$ and degree d can be defined as

$$h_d(x_1, ..., x_n) := \sum_{1 \leq i_1 \leq ... \leq i_d \leq n} x_{i_1} ... x_{i_d},$$

thus for instance

$$h_0(x_1, ..., x_n) = 0,$$
$$h_1(x_1, ..., x_n) = x_1 + ... + x_n,$$

and

$$h_2(x_1, ..., x_n) = x_1^2 + ... + x_n^2 + \sum_{1 \leq i < j \leq n} x_i x_j.$$

One can also define all the complete homogeneous symmetric polynomials of n variables simultaneously by means of the generating function

$$\sum_{d=0}^{\infty} h_d(x_1, ..., x_n) t^d = \frac{1}{(1-tx_1)...(1-tx_n)}. \tag{1}$$

We will think of the variables $x_1, ..., x_n$ as taking values in the real numbers. When one does so, one might observe that the degree two polynomial h_2 is a positive definite quadratic form, since it has the sum of squares representation

$$h_2(x_1, ..., x_n) = \frac{1}{2} \sum_{i=1}^n x_i^2 + \frac{1}{2}(x_1 + ... + x_n)^2.$$

In particular, $h_2(x_1, ..., x_n) > 0$ unless $x_1 = ... = x_n = 0$. This can be compared against the superficially similar quadratic form

$$x_1^2 + ... + x_n^2 + \sum_{1 \leq i < j \leq n} \epsilon_{ij} x_i x_j$$

where $\epsilon_{ij} = \pm 1$ are independent randomly chosen signs. The *Wigner semicircle law* says that for large n, the eigenvalues of this form will be mostly distributed in the interval $[-\sqrt{n}, \sqrt{n}]$ using the semicircle distribution, so in particular the form is quite far from being positive definite despite the presence of the first n positive terms. Thus the positive definiteness is coming from the finer algebraic structure of h_2, and not just from the magnitudes of its coefficients.

One could ask whether the same positivity holds for other degrees d than two. For odd degrees, the answer is clearly no, since $h_d(-x_1, ..., -x_n) = -h_d(x_1, ..., x_n)$ in that case. But one could hope for instance that

$$h_4(x_1,...,x_n) = \sum_{1 \leq i \leq j \leq k \leq l \leq n} x_i x_j x_k x_l$$

also has a sum of squares representation that demonstrates positive definiteness. This turns out to be true, but is remarkably tedious to establish directly. Nevertheless, we have a nice result of Hunter that gives positive definiteness for all even degrees d. In fact, a modification of his argument gives a little bit more:

Theorem 1 Let $n \geq 1$, let $d \geq 0$ be even, and let $x_1,...,x_n$ be reals.

- (i) (Positive definiteness) One has $h_d(x_1,...,x_n) \geq 0$, with strict inequality unless $x_1 = ... = x_n = 0$.
- (ii) (Schur convexity) One has $h_d(x_1,...,x_n) \geq h_d(y_1,...,y_n)$ whenever $(x_1,...,x_n)$ majorises $(y_1,...,y_n)$, with equality if and only if $(y_1,...,y_n)$ is a permutation of $(x_1,...,x_n)$.
- (iii) (Schur-Ostrowski criterion for Schur convexity) For any $1 \leq i < j \leq n$, one has $(x_i - x_j)(\frac{\partial}{\partial x_i} - \frac{\partial}{\partial x_j}) h_d(x_1,...,x_n) \geq 0$, with strict inequality unless $x_i = x_j$.

Proof: We induct on d (allowing n to be arbitrary). The claim is trivially true for $d = 0$, and easily verified for $d = 2$, so suppose that $d \geq 4$ and the claims (i), (ii), (iii) have already been proven for $d-2$ (and for arbitrary n).

If we apply the differential operator $(x_i - x_j)(\frac{\partial}{\partial x_i} - \frac{\partial}{\partial x_j})$ to $\frac{1}{(1-tx_1)...(1-tx_n)}$ using the product rule, one obtains after a brief calculation

$$\frac{(x_i - x_j)^2 t^2}{(1-tx_1)...(1-tx_n)(1-tx_i)(1-tx_j)}.$$

Using (1) and extracting the t^d coefficient, we obtain the identity

$$(x_i - x_j)(\frac{\partial}{\partial x_i} - \frac{\partial}{\partial x_j}) h_d(x_1,...,x_n)$$
$$= (x_i - x_j)^2 h_{d-2}(x_1,...,x_n,x_i,x_j). \quad (2)$$

The claim (iii) then follows from (i) and the induction hypothesis.

To obtain (ii), we use the more general statement (known as the *Schur-Ostrowski criterion*) that (ii) is implied from (iii) if we replace h_d by an arbitrary symmetric, continuously differentiable function. To establish this criterion, we induct on n (this argument can be made independently of the existing induction on d). If $(y_1,...,y_n)$ is majorised by $(x_1,...,x_n)$, it lies in the permutahedron of $(x_1,...,x_n)$. If $(y_1,...,y_n)$ lies on a face of this permutahedron, then after permuting both the x_i and y_j we may assume that $(y_1,...,y_m)$ is majorised by $(x_1,...,x_m)$, and $(y_{m+1},...,y_n)$ is majorised by $(x_{m+1},...,x_n)$ for some $1 \leq m < n$, and the claim then follows from two applications of the induction hypothesis. If instead $(y_1,...,y_n)$ lies in the interior of the permutahedron, one can follow it to the boundary by using one of the vector fields $(x_i - x_j)(\frac{\partial}{\partial x_i} - \frac{\partial}{\partial x_j})$, and the claim follows from the boundary case.

Finally, to obtain (i), we observe that $(x_1,...,x_n)$ majorises $(x,...,x)$, where x is the arithmetic mean of $x_1,...,x_n$. But $h_d(x,...,x)$ is clearly a positive multiple of x^d, and the claim now follows from (ii). □

If the variables $x_1,...,x_n$ are restricted to be nonnegative, the same argument gives Schur convexity for odd degrees also.

The proof in Hunter of positive definiteness is arranged a little differently than the one above, but still relies ultimately on the identity (2). I wonder if there is a genuinely different way to establish positive definiteness that does not go through this identity.

Comments (19)

unowatblog · 6 August 2017

There is a typo in the definition of h_d: the last index is i_d rather than i_n.
Aside, since Schur convexity seems to me a natural solution – would you consider sum of squares type identity a different solution?

> **Terence Tao** · 7 August 2017
>
> Thanks for the correction!
> One could convert the Schur convexity proof into a (rather artificial looking) sum of squares proof, but if there was a nice looking sum of squares representation analogous to that for h_2 then I would consider that to be a different proof.
> Here is one small observation that might help towards a different proof: if one already knows the non-negativity of $h_d(x_1, ..., x_n)$ for even d, then this implies the non-negativity of $h_d(x_1, ..., x_n, y, -y)$ for any y and even d. This is because the generating function $\sum_{d=0}^\infty h_d(x_1, ..., x_n, y, -y)t^d$ is equal to the generating function of $\sum_{d=0}^\infty h_d(x_1, ..., x_n)t^d$ times $\frac{1}{(1-yt)(1+yt)} = 1 + y^2t^2 + y^4t^4 + ...$; the latter has only even powers and non-negative coefficients, hence the claim. So one can "cancel" any pair $y, -y$ that occurs in $x_1, ..., x_n$, which for instance demonstrates non-negativity of $h_d(x_1, ..., x_n)$ whenever the $x_1, ..., x_n$ all have the same magnitude. I wasn't able to convert this argument into a proof of the general case though.

andrescaicedo · 6 August 2017

At least on my screen the tag (2) is not visible. (I found it by looking at the TeX code of the displayed identity.)
[Corrected, thanks – T.]

Orr Shalit · 6 August 2017

It might be better to start the induction with d=2 (to obtain the condition for strict inequality).
[Suggestion implemented, thanks – T.]

Anonymous · 7 August 2017

Maybe this works. Let $Z_1, ..., Z_n$ be i.i.d exponentially distributed with parameter 1. Let $S = (\sum_{i=1}^n x_i Z_i)^d/d!$. The coefficient of a monomial $\prod_i x_i^{d_i}$ in S is $\prod_i Z_i^{d_i}/d_{(i)!}$ whose expected value is 1 because the d_i'th moment of Z_i is $d_{(i)!}$. So $h_d(x_1, ..., x_n)$ is the expected value of S, clearly non-negative for even d.

> **Synia** · 7 August 2017
>
> Really nice ! Another way of seeing that $h_d(x_1, ..., x_n) = \mathbb{E}\left(\frac{1}{d!}(\sum_{j=1}^n x_j Z_j)^d\right)$ is by writing $h_d(x_1, ..., x_n) = [t^d]\prod_{j=1}^n \frac{1}{1-x_j t} = [t^d]\prod_{j=1}^n \mathbb{E}\left(e^{tx_j Z_j}\right)$
> $= [t^d]\mathbb{E}\left(e^{t\sum_{j=1}^n x_j Z_j}\right)$ where $[t^d]f(t)$ is the d-th Fourier coefficient (here, we suppose $|tx_j| < 1$ for all j for the convergence).
>
> **Terence Tao** · 7 August 2017
>
> Very nice! I was hunting for a probabilistic interpretation of h_d (this being another major way to prove positivity results, besides sum of squares methods and induction methods) but did not see this very neat moment interpretation. I think this is a good demonstration of the power of the probabilistic method.

(It now makes me wonder more generally if Schur polynomials have a similarly nice probabilistic interpretation. Of course, one can simply plug in the probabilistic interpretation of h_d into the first Jacobi-Trudi formula, but I'm hoping for something more than this...)



Aditya Guha Roy · 15 September 2021

In fact Hunter's result says that $h_{2r}(x_1, ..., x_n) \geq \frac{||x||^{2r}}{2^r r!}$; I strongly believe that this probabilistic argument can be amplified to deduce Hunter's original inequality too.



David Speyer · 8 August 2017

Very nice indeed!
I would say that this is morally a "sum of squares" argument; you have expressed h_d as an integral of a d-th power. I imagine that, by replacing the integral by a suitable average over finitely many values, you could obtain a literal sum of squares expression.



Terence Tao · 8 August 2017

True; indeed it looks like all one needs to do is to replace the exponential random variable Z by a discrete random variable which matches moments with the continuous one up to d^{th} order and one would get a discrete representation. (The existence of such a discrete random variable follows from Caratheodory's theorem.)



Arash B · 8 August 2017

"To establish this criterion, we again induct on d "; Isn't it induction on n?
[Corrected, thanks – T.]



David Speyer · 8 August 2017

You might already know this, but there is a positivity property for Schur polynomials: If λ is a partition with all even parts, then $s_\lambda(x_1, ..., x_n)$ is nonnegative for any real inputs. (And, I think, positive unless the inputs are all zero, but I'm not sure.)
By symmetry and continuity, it is enough to show that $s_\lambda(x_1, ..., x_n) > 0$ when $x_1 > x_2 > \cdots x_n$. We recall the ratio of alternants formula: $s_\lambda(x_1, ..., x_n) = \det(x_i^{\lambda_j+n-j})/\det(x_i^{n-j})$. So it is enough to show that $\det(x_i^{\lambda_j+n-j}) > 0$ when $x_1 > x_2 > \cdots x_n$ and all the λ_i are even. By continuity, it is enough to show that $\det(x_i^{\lambda_j+n-j}) \neq 0$ under these conditions.
Suppose to the contrary that $\det(x_i^{\lambda_j+n-j}) = 0$. Then there is a linear relation $\sum_j c_j x_i^{\lambda_j+n-j} = 0$. In other words, the polynomial $\sum c_j x^{\lambda_j+n-j}$ has n distinct real roots. But the condition that the parts of λ are even means that that the exponents $\lambda_j + n - j$ alternate in parity, which means Descartes' Rule of Signs will permit only $n-1$ real roots, a contradiction. $\square$
If we wanted to show strict positivity, we'd need to show that $x_i - x_j$ only divided $\det(x_i^{\lambda_j+n-j}) \neq 0$ once; we've shown strict positivity off the locus where two of the x_i are equal already.
I believe this was an exercise in Barvinok's convexity textbook, but I don't have it at hand to check.



Synia · 9 August 2017

This gives an alternative proof for the positivity of h_{2n} since $h_n = s_{(n)}$.



Terence Tao · 9 August 2017

Nice! (I knew one could use the rule of signs to prove that the Schur polynomials were positive when the inputs $x_1, ..., x_n$ were positive, but this was already obvious from the Young tableau expansion.)
Unfortunately most Schor polynomials can vanish outside of the origin; indeed, by testing a Schur polynomial s_λ on a basis vector $(1, 0, ..., 0)$ using the Young tableau expansion, one gets zero

unless λ has only one non-zero part. So the only strictly positive definite Schur polynomials are the h_{2n}. (Apoorva Khare and I have an upcoming paper where we had a theorem that worked for all strictly positive definite Schur polynomials, so it was a bit disappointing to realise how few of them there were.)



David Speyer · 10 August 2017

Oh wow, I should have noticed that about $s_\lambda(1,0,0,\ldots,0) = 0$. Huh, I wonder what the real points where s_λ vanishes, when all parts of λ are even, look like.



Anonymous · 21 August 2017

Theorem 1(i) is easy if all the reals have the same sign. Otherwise, without loss of generality, x_1 is positive and x_2 is negative. The identity

$$\frac{1}{1-tx_1}\frac{1}{1-tx_2} = \frac{x_1}{x_1+(-x_2)}\frac{1}{1-tx_1} + \frac{(-x_2)}{x_1+(-x_2)}\frac{1}{1-tx_2},$$

shows that $h_d(x_1,x_2,\ldots,x_n)$ is a convex combination of $h_d(x_1,0,\ldots,x_n)$ and $h_d(0,x_2,\ldots,x_n)$. These are each strictly positive for even d by induction on the number of non-zero components.



Anonymous · 23 August 2017

Here is a proof of strict positivity, using log-convexity.

This approach gives Hunter's bound

$h_{2r} \geq (\sum x_i^2)^r/2^r r!$, which I don't see a way to get just using moment generating functions.

Define $\mathcal{C}$ to be the set of exponential generating functions of log-convex sequences.

The Davenport-Pólya theorem says that $\mathcal{C}$ is closed under products.

I will use the following lemma. If $f, g \in \mathcal{C}$ then the even coefficients of $f(t)g(-t)$ are non-negative.

Proof: the coefficient of t^n is $\sum_i \binom{n}{i}(-1)^i f_i g_{n-i}$.

The sequence $f_0 g_n, f_1 g_{n-1}, \ldots, f_n g_0$ is log-convex, which implies it is convex.

But a sum of the form $\sum_i \binom{n}{i}(-1)^i c_i$ is non-negative if n is even and $c_0,\ldots,c_n$ is convex. QED.

Assume $x_1,\ldots,x_m > 0$ and $0 > x_{m+1},\ldots,x_n$. By abuse of notation, write $h_d = h_d(x_1,\ldots,x_n)$, and $h_d^+ = h_d(x_1,\ldots,x_m)$, and $h_d^- = h_d(-x_{m+1},\ldots,-x_n)$. We have the identity $\sum_{d\geq 0} h_d t^d = (\sum_{d\geq 0} h_d^+ t^d)(\sum_{d\geq 0} h_d^-(-t)^d)$. Note $\mathcal{C}$ contains all functions of the form $1/(1-tx)$ for $x>0$, and so contains $\sum_{d\geq 0} h_d^+ t^d$ and $\sum_{d\geq 0} h_d^- t^d$ by the formula (1) in the main post. By the lemma above, $h_d \geq 0$ for even d.

To get a bound in terms of $\sum x_i^2$, note that $\exp(-x^2t^2)/(1-tx)$ is

in $\mathcal{C}$ for $x>0$:

the first few coefficients of $(tx)^k/k!$ are

[A000266](https://oeis.org/A000266), and they tend quickly to $k!e^{-1/2}(1+o(1))$ which is log-convex.

By the same argument as before, the even coefficients of

$\exp(-\sum x_i^2 t^2/2)\sum_{d\geq 0} h_d t^d$

are non-negative. Multiplying by $\exp(\sum x_i^2 t^2/2)$

gives the bound $h_{2r} \geq (\sum x_i^2)^r/2^r r!$.



Anonymous · 23 August 2017

Sorry, I realised that proof doesn't work – convexity by itself isn't strong enough to guarantee those even coefficients are non-negative in general.



Silouanos_Brazitikos · 16 January 2026

Dear Terry,

Here is a recent sharp lower bound for (CHP), confirming Hunter's conjecture.

https://arxiv.org/abs/2512.12254